Al-Balqa Applied University, Faculty of Engineering Technology, Basic Engineering Sciences, Fall Semester 2022/2023

Numerical Methods

Mid Term Exam

19/30

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Mark:

Student's Name:

ID Number: .....

Date of Exam: 27-11-2022 Sunday (14:30-15:20)

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Question1: Solve the first two iterations to find the root of $f(x) = -1 + 5.5x - 4x^2 + 0.5x^3$ *and the* $x_o = 0.25$ using Newton Raphson Method.

(2)

n	X_i	$f(X_i)$	$f'(X_i)$	X_{i+1}
1	0.25	0.13281	3.59375	0.21304
2	0.21304	-0.00498	3.86375	0.21432

(2) (2) (2)

$f'(x) = \frac{3}{2}x^2 - 2x + 5.5$

$X_{i+1} = X_i - \frac{f(x_i)}{f'(x_i)}$

$X_r = 0.21432$

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Question2: Based on the bisection method, find the root of $f(x) = 3x^2 - e^x$ that is located in [3.5,4] with error less than $\varepsilon = 0.1$

n	X_L	X_u	$f(x_L)$	$f(X_u)$	X_r	$f(X_r)$	ε_a %
1	3.5	4	+	-	3.75	-	-
2	3.75	4	-	-	3.875	-	0.125
3	3.875	4	-	-	3.9375	-	0.0625

< 0.1

$X_r = 3.9375$

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Question3: Use Gauss elimination to find α where $x_1 = 3$

$$30\alpha x_1 - \alpha x_2 - 2\alpha x_3 = 7.8$$
$$\alpha x_1 + 70\alpha x_2 - 3\alpha x_3 = -19.3$$
$$3\alpha x_1 - 2\alpha x_2 + 100\alpha x_3 = 71.4$$



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$$Ax = b$$
$$\downarrow$$
$$L \cdot U \cdot x = b \qquad L \cdot U = A$$
$$L \cdot d = b$$

4 **Question5:** if $Ax = b$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ 0.375 & 0.23913 & 1 \end{bmatrix}, \text{ and } [U] = \begin{bmatrix} -8 & 1 & -2 \\ 0 & -5.75 & -1.5 \\ 0 & 0 & 8.108696 \end{bmatrix}, \quad b = \begin{bmatrix} -20 \\ -38 \\ -34 \end{bmatrix}$$

If $[b]^T = [-20 \quad -38 \quad -34]$, find: (show details of solution)

1. [A]
2. x_3?

① $|A| = \begin{bmatrix} 1 & 0 & 0 \\ -0.25 & 1 & 0 \\ 0.375 & 0.23913 & 1 \end{bmatrix} \cdot \begin{bmatrix} -8 & 1 & -2 \\ 0 & -5.75 & -1.5 \\ 0 & 0 & 8.108696 \end{bmatrix}$

$a_{11} = 1 * -8 + 0 + 0 = -8$

$a_{12} = 1 + 0 + 0 = 1$

$a_{13} = 1 * -2 + 0 + 0 = -2$

$a_{21} = 2 + 0 + 0 = 2$

$a_{22} = -0.25 - 5.75 + 0 = -6$

$a_{23} = 0.5 - 1.5 + 0 = -1$

$a_{31} = 0.375 \times 8 = -3$

$a_{32} = (0.375 \times 1) + (0.23913 \times -6) = -0.4999975$

$a_{33} = (0.375 \times -2) + (0.23913 \times -1) + (1 \times 8.108696) = 7$

$$[A] = \begin{bmatrix} -8 & 1 & -2 \\ 2 & -6 & -1 \\ -3 & -1 & 7 \end{bmatrix}$$

~~$L \cdot d = b$~~

$$R_2 + \frac{R_1}{4}, \quad R_3 + \frac{3}{8}R_1 \begin{bmatrix} -8 & 1 & -2 \\ 2 & -6 & -1 \\ -3 & -1 & 7 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -20 \\ -38 \\ -34 \end{bmatrix}$$

$$\downarrow$$

$$\left[\begin{array}{ccc|c} -8 & 1 & -2 & -20 \\ 0 & -5.75 & -1.5 & -43 \\ 0 & -\frac{11}{8} & 7.75 & -26.5 \end{array}\right] \quad R_3 + \frac{5}{4}R_2$$

$$\left[\begin{array}{ccc|c} -8 & 1 & -2 & -2 \\ 0 & -5.75 & -1.5 & -43 \\ 0 & 0 & 7.375 & -37.25 \end{array}\right]$$

$$7.375 X_3 = -37.25$$

$$X_3 = -5.05$$

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Question4: Find the value β that the system $\begin{bmatrix} 1 & 1 & -1 \\ 2 & \beta & 1 \\ 3 & 2 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ -4 \end{bmatrix}$ so that no inverse exists (Singular).

$\det A = 0$

$$\det A = 1\begin{vmatrix} \beta & 1 \\ 2 & -4 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ 3 & -4 \end{vmatrix} + (-1)\begin{vmatrix} 2 & \beta \\ 3 & 2 \end{vmatrix} = 0$$

$$(-4\beta - 2) + 11 - 1(4 - 3\beta) = 0$$

$$-4\beta - 2 + 11 - 4 + 3\beta = 0$$

$$-\beta + 5 = 0$$

$$\frac{-\beta}{-1} = \frac{-5}{-1}$$

$$\boxed{\beta = 5}$$

Check:

$$-22 + 11 + 11 \stackrel{?}{=} 0$$

$$0 \stackrel{?}{=} 0 \quad \checkmark$$

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